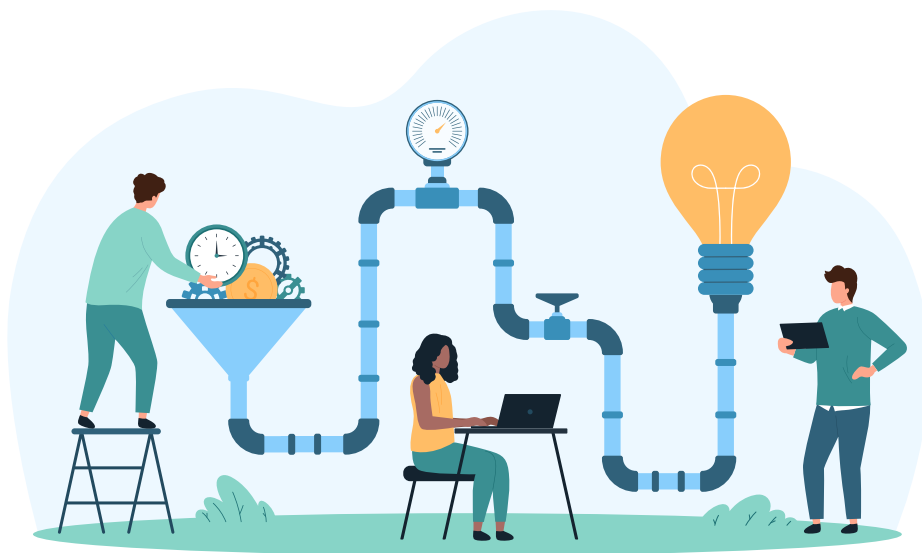


Using Kafka Connectors for Data Pipelines: An Overview

Find out how to use Kafka Connect and the pre-built connectors that come with it to connect Kafka to the external systems that businesses use, including databases and cloud storage. There are some best practices for using these connectors too.



These days, it's hard to imagine any serious business not needing real-time data. Whether it's live metrics from devices, syncing updates from databases, or tracking users as they click around an app, moving and handling data instantly has become non-negotiable.

Apache Kafka plays a major role here as it powers countless real-time systems. It's fast, fault-tolerant, and handles crazy amounts of data. But while Kafka does a great job moving data, it doesn't automatically connect to all the tools businesses use like databases, cloud storage, or search platforms.

That's where Kafka Connect comes in as it acts like a bridge. With it, you can hook Kafka up to all sorts of external systems and don't need to write a bunch of glue code to do it. You can just use pre-built connectors and get your data flowing in and out quickly.

Kafka connectors are pluggable components of the Kafka Connect system that facilitate effortless data integration between Kafka and other systems. There are two types: source connectors, which extract data from systems such as databases or message queues and put it into Kafka topics, and sink connectors, which send data from Kafka to targets such as data warehouses, search indexes, or file systems. By removing the requirement for custom ingestion code, Kafka connectors make it easier to implement scalable, fault-tolerant pipelines in real-time and batch processing pipelines.

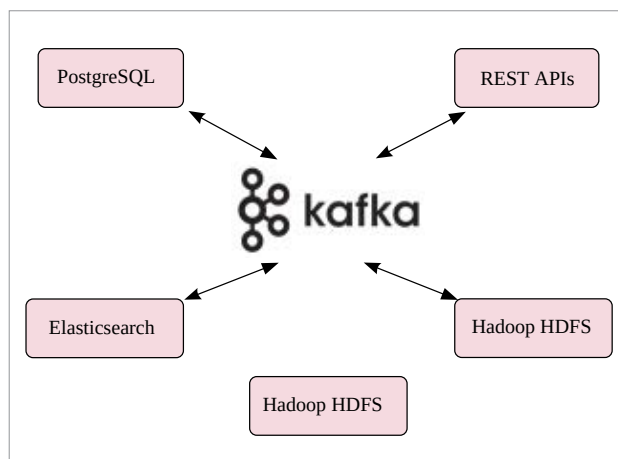


Figure 1: Kafka Connect ecosystem

Anatomy of the Kafka data pipeline

Key components of the Kafka-driven pipeline: In a Kafka data pipeline, several important components work together to handle real-time streaming of data. The Kafka broker is an important component of the system that helps make it scalable and fault-tolerant by receiving and processing messages based on topics and partitions. Producers write Kafka, and consumers read it. To integrate Kafka with external systems, Kafka Connect is used to deploy connectors on workers